

BENJAMIN RIOT-BRETÊCHER

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EDUCATION

Ph.D - Atmospheric and Oceanic Sciences, McGill University 2022 -

Supervisor: [Prof. Yi Huang](#)

Relevant Coursework: Atmospheric Dynamics, Atmospheric Radiation, Numerical Methods, Waves & Stability

ACHIEVEMENTS: Comprehensive Examination (February 2024); Proposal Examination (August 2025)

B.S - Atmospheric and Oceanic Sciences, Université du Québec à Montréal 2019 - 2022

PROFESSIONAL EXPERIENCE

Teaching Assistant 2023 -

McGill University

- [Introduction to Atmospheric Sciences - ATOC181](#) (Fall 2023, Fall 2025)
- [Science of Storms - ATOC184](#) (Winter 2024, Winter 2026)

Visitor Student Oct 2024 - Jan 2025

[School of Atmospheric Sciences, Nanjing University](#)

Treasurer Aug 2023 - Aug 2024

Council of Atmospheric and Oceanic Sciences (CAOS)

Field Campaign Involvement 2022 - Present

McGill University

- [ECALOT Field Campaign](#) (Fall 2024)
- [WHAFFERS Field Campaign](#) (Winter 2025)
- [GHG Montreal Project](#) (Ongoing)
- [PONEX](#) (Winter 2026)

Research Assistant May 2022 - Aug 2022

McGill University

Supervisor: [Prof. Yi Huang](#)

Research Assistant June 2021 - Aug 2021

Université du Québec à Montréal

Supervisor: [Prof. Jean-Pierre Blanchet](#)

PUBLICATIONS

- **B. Riot-Bretêcher**, Y. Huang, L. Liu, J. Gero, [Strengthening Atmospheric Greenhouse Effect at the Arctic North Slope of Alaska Site Evidenced by Long-Term Records of the Downwelling Longwave Radiance Spectrum](#), Journal of Geophysical Research: Atmospheres, 2025.
- Y. Wang, G. X. He, **B. Riot-Bretêcher**, M. Pelletier, Y. Huang, S. K. Loeb, *Integrating Meteorological Data into Linear Mixed Models for Sunlight Inactivation of Indicator Organisms in Natural Lake Water* (in prep)
- B. Pery, H. Brindley, J. Murray, R. Hogan, S. Mosselmans, S. Panditharatne, L. Warwick, T. L'Ecuyer, **B. Riot-Bretêcher**, Y. Huang, *Towards ground-to-space radiative closure in the mid- to far-infrared* (in prep)
- Z. Qu et al., **B. Riot-Bretêcher**, *EarthCARE Commissioning Calibration/Validation Campaign in Ottawa (ECALOT)* (in prep)

CONFERENCES AND TALKS

- [Joint FORUM-HAWC Workshop - Poster](#) - Montreal, QC, Canada - October 2025
- [Gordon Research Conference on Radiation and Climate - Invited Speaker](#) - Lewiston, ME, USA - July 2025
- [Gordon Research Seminar on Radiation and Climate - Speaker](#) - Lewiston, ME, USA - July 2025
- [2nd ESA-JAXA EarthCARE In-Orbit Validation Workshop - Co-author, Presented by L. Liu](#) - Frascati, Italy - March 2025 Validation of EarthCARE cloud retrievals using surface spectral infrared radiances from the ECALOT campaign. (Liu, L., **Riot-Bretêcher**, B. et al.)

- [Nanjing University - Atmospheric Sciences Seminar](#) - **Speaker** - Nanjing, China - December 2024
- [43rd Canadian Symposium on Remote Sensing](#) - **Attendee** - Québec City, QC, Canada - July 2022
- [Canadian Meteorological and Oceanographic Society](#) - **Attendee** - Remote - June 2021

AWARDS AND SCHOLARSHIP

- McGill Graduate Excellence Fellowship (2022 - 2028)
- Graduate Mobility Award - McGill University (Oct 2024 - Jan 2025) - \$ 3,800
- Canadian Meteorological and Oceanographic Society Undergraduate Scholarship (April 2021) - \$ 1,000

SKILLS

Technical Skills	MATLAB, Fortran, Python, Java, R, Bash, Linux, Radiative Transfer Model, Weather Balloon Launches, AERI Maintenance, Optimal Estimation Method
Mathematical Skills	Algebra, Calculus, Probability, Statistics
Certifications	• Data Security - Compute Ontario (June 2025) • Introduction to Advanced Research Computing - Compute Ontario (June 2025) • Introduction to Version Control Using Git - Compute Ontario (June 2025) • Scaling Up HPC Workflows - Compute Ontario (June 2025) • Career Essentials in Software Development by Microsoft and LinkedIn (Certificate ID) • Programming Foundation: Beyond the Fundamentals by LinkedIn (Certificate ID)